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11 - dars: Ensemble Methods

Ensemble – birqalikda, jamoaviy, jō'r bō'lish degani.

1. ML technique
2. Bir nechta kuchsiz modellarni birlashtirish orqali ~~modellarni~~ modelni rivojlantirish
3. Asosiy logika: Ko'plab kuchsiz algorithm/kuchli model xosil qiladi.

Masalan Jahon chempionatiya terma qilib kuchsiz futbolchilarnda qaysidir kuchsiz joylarini to'ldirish u-n bitta jamoa qilib bir-birini to'ldiruvchi kuchli jamoa hosil qilishdek gap.

Nima uchun ishlatamiz?

1. Reduce variance → handle overfitting (e.g., bagging)
2. Reduce bias → improve weak learners (e.g., boosting)
   tarafkashlik
3. Improve accuracy → aggregate multiple perspectives (Turli hil xolatlarda tekshirib ko'rib albatta natijani rivojlantirib beradi)
4. Increase robustness → less sensitive to noise (Datasetinglarni kengaytirganingdan keyin model stabil ishlashda davom etishi kerak chunki nana misol u-n biz. Model yaratdik deylik, model

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yaratganimizdan keyin uni biror website ga qo'yib uni website ni ag'illiklashtirib berib, u website dan har kuni bir xil user'lar foydalanmaydida. Masalan ba'zi saytlarda qo'shilar ko'p bo'ladi sababi yuklama ko'p bo'ladi, ko'p odamlar qatnashgani u-n. Huddi shunaqa holatlarda Agar biz = modelimiz Robustness bo'lmagan bo'lsa stabil bo'lmaydi.

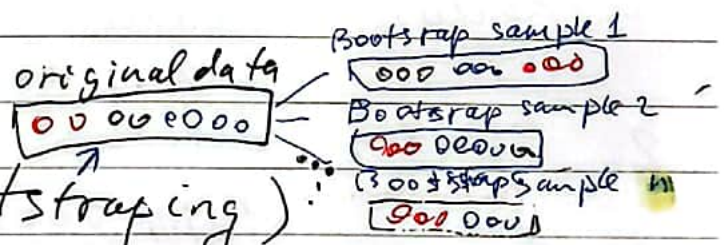
Masalan: Sayasetimiz 5000 ta user birdaniga kelib qolsa, 10000000 ta kelib qolsa... nima bo'ladi?

Huddi shunday holatlariga sensitive bo'lishi k-k.

Ensemble Methods: Turlari

1. Bagging (Boostraping) — method; Bitta fenomen
2. Boosting
3. Stacking
4. Voting

Eng ko'p foydalaniladigan Ensemble method'imiz



- Bagging (Bootstraping):

~~1. Train multiple~~

birinchi qoida: Dataset'ni length'i o'zgarmaydi, tushib qolishi mumkin va qaytalanishi mumkin va tashqaridan biror element kiritilmaydi. Ichki resurslardan foydalangan holda foydalanamiz

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, bunda turli xil qiymatlarga ajratamiz

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Misol: [1, 2, 3], ni

① [3, 2, 1], ② [2, 1, 3], ③ [1, 3, 2], ④ [1, 1, 2]

⑤ [1, 2, 1], ⑥ [1, 3, 1], ⑦ [1, 1, 1], ⑧ [2, 2, 2]

....

1. Bir xil len, tushib qolish va qaytalanish normal va tashqaridan birorta element kirmaydi faqat - va - faqat ichki resursdan foydalanamiz

2. Train multiple: bu yuqoridagi misollarni har birini train qiladi. Tasavvur qiling har birini bitta Decision Tree deb.

3. Asosiy algoritmimiz: Random Forest bóladi.

Formula: Final Accuracy (Avg of all iteration) = 92%

4. Result: Voting / avg (Majority Voting (classification) Regression (Average))

Errorni ham shunday topadi. R2-scoreni ham shunday topadi.

Bagging methodini ichida birinchi Bootstrapping orqali turli xil sample/ga (qism/ga) ajratib olayapmiz.

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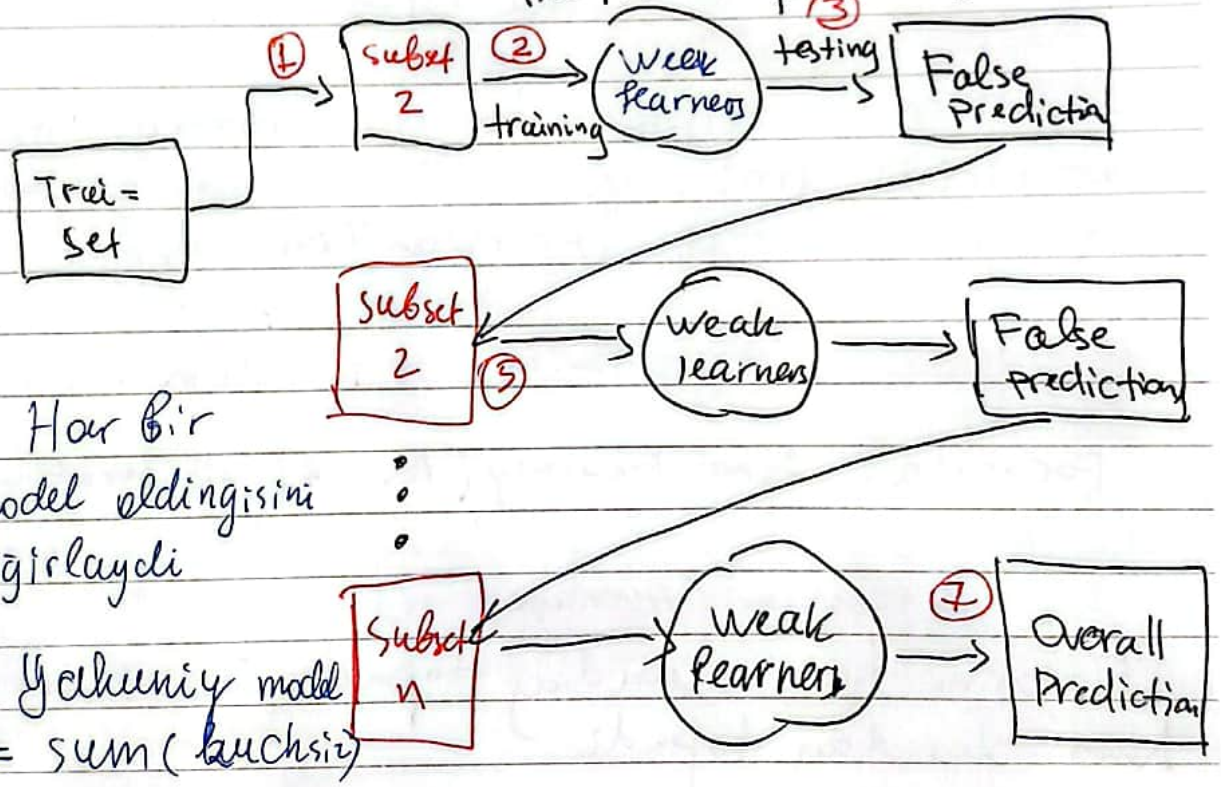
- Boosting:

1. Train ketma-ket (Bunda Baggingga o'xshash dataset/ni alohida-alohida qismlarga ajratmaymiz)

2. Full dataset b/n bajaradi

The Process of Boosting



3. Har bir model oldingisini tog'irlaydi

4. Yakuniy model = sum(buchsiz)

5. Algorithm: GBM, Adaboost, XGB lar bor ya'ni XGB ichida ~~bir nechta~~ turli xil algoritm bor u bir-birini rivojlantirib teng bitta kuchli algoritm xosil qiladi. va kuchli natija hosil qilib beradi.

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Bagging

```
from sklearn.model_selection import train_test_split

X = df.drop('xG', axis=1)
y = df['xG']

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size = 0.2, random_state = 42)

# Random Forest
from sklearn.ensemble import RandomForestRegressor
model = RandomForestRegressor()

# train qilamiz
model.fit(X_~~test~~train, y_train)

# prediction toparmiz.
y_pred = model.predict(X_test)

# Metrics Evaluation
from sklearn.metrics import r2_score, mean_absolute_error

r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)

# Print qilish
print('R2_SCORE:', r2)
print('MEAN_ABSOLUTE_ERROR:', mae)
```

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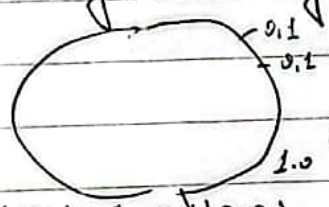
Boosting (XG Boost)

pip install xgboost

import xgboost as xgb

model = xgb.XGBRegressor(n_estimators = 100,
learning_rate = 0.1, max_depth = 3, subsample=1,
O'rganish oraliqi colsample_bytree = 0.8, random_
state = 42)



~~agar biz~~ #yar biz katta qilib yuborsak learning rateni oraliq masqun kattalashib ketadi.

model.fit(X_train, y_train)

# predict
y_pred = model.predict(X_test)